National Institute of Technology Kurukshetra

Department of Computer Engineering

Problem Solving & Programming Skills II (CSIC 103)

Branch: Robotics & Automation (R&A)

Date of Exam – 24-October-2024

Time: 02:30 PM-03:20 PM [50 Min]

Max Marks:20

1 Write a C program that uses a function to check whether a given positive integer is a perfect number using. **6M**

According to Wikipedia: In number theory, a perfect number is a positive integer that is equal to the sum of its proper positive divisors, that is, the sum of its positive divisors excluding the number itself (also known as its aliquot sum). Equivalently, a perfect number is a number that is half the sum of all of its positive divisors (including itself).

Example: The first perfect number is 6, because 1, 2, and 3 are its proper positive divisors, and 1 + 2 + 3 = 6. Equivalently, the number 6 is equal to half the sum of all its positive divisors: (1 + 2 + 3 + 6) / 2 = 6. The next perfect number is 28 = 1 + 2 + 4 + 7 + 14. This is followed by the perfect numbers 496 and 8128.

Expected Input:

Enter an integer: 6

Expected Output:

True

2 What would be the output of the following code snippets? Explain your answer. **[3+5]M**

a.

```
#include <stdio.h>
int x=5;
void myFunction() {
  int x=6;
  printf("%d\n", ++x);
}
void main() {
  myFunction();
  printf("%d\n", x);
}
```

b.

```
#include <stdio.h>
void fun(int n) {
if (n > 0) {
    fun(n- 1);
    printf("%d ", n);
    fun(n- 1);
}}
void main() {
    int num = 4;
    fun(num);
}
```

3 Consider a 2D array arr[][] of size 4 x 4. Write a C program that uses **for loops** to calculate the sum and average of the elements in the subarray defined by the top-left corner (0,0) and the bottom-right corner (1,1). The program should take the elements of the array as input from the user, compute the sum of the specified subarray, and display both the sum and the average (formatted to two decimal places). **6M**

Expected Input:

The program should prompt the user to input 16 integers to populate a 4x4 array.

Expected Output:

Display the elements of the subarray from (0,0) to (1,1).

Compute and display the sum of these elements.

Finally, display the average of the subarray.

121 3121 4121 3

121